Island of inversion around ^{64}Cr

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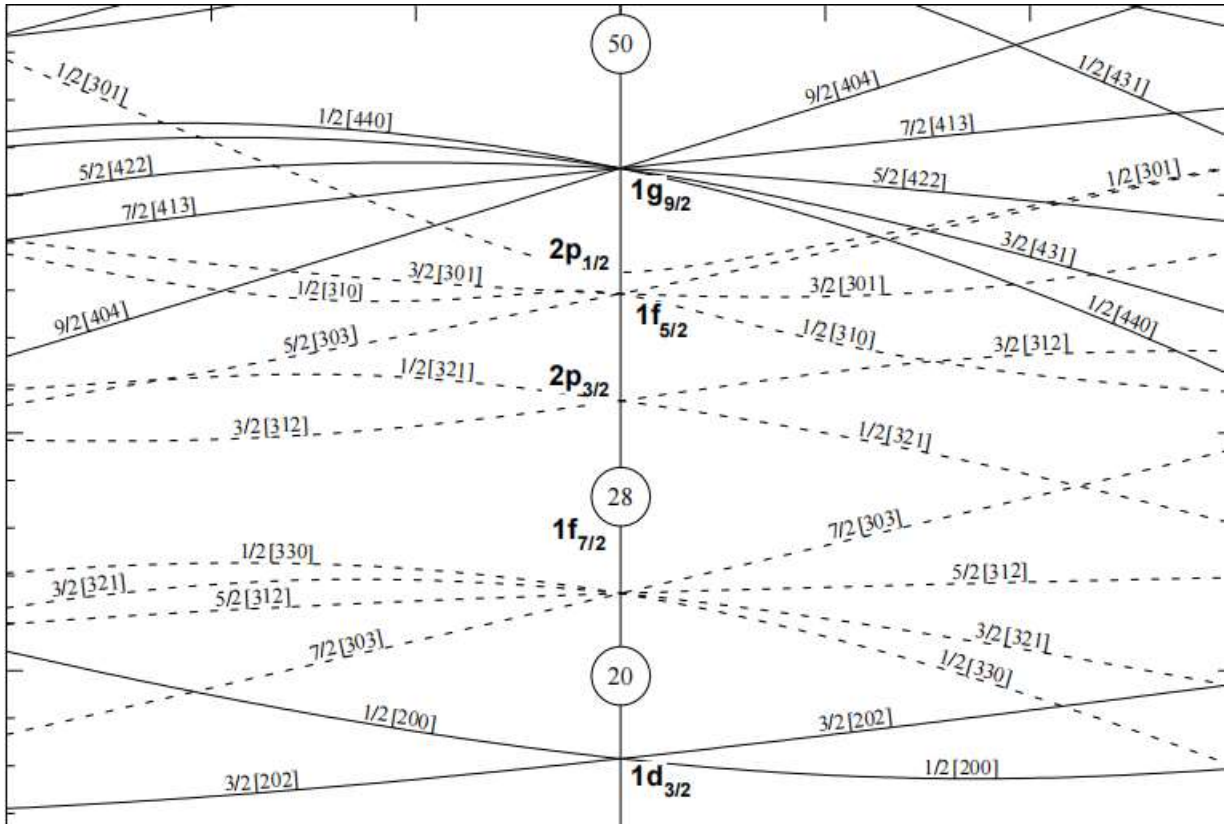
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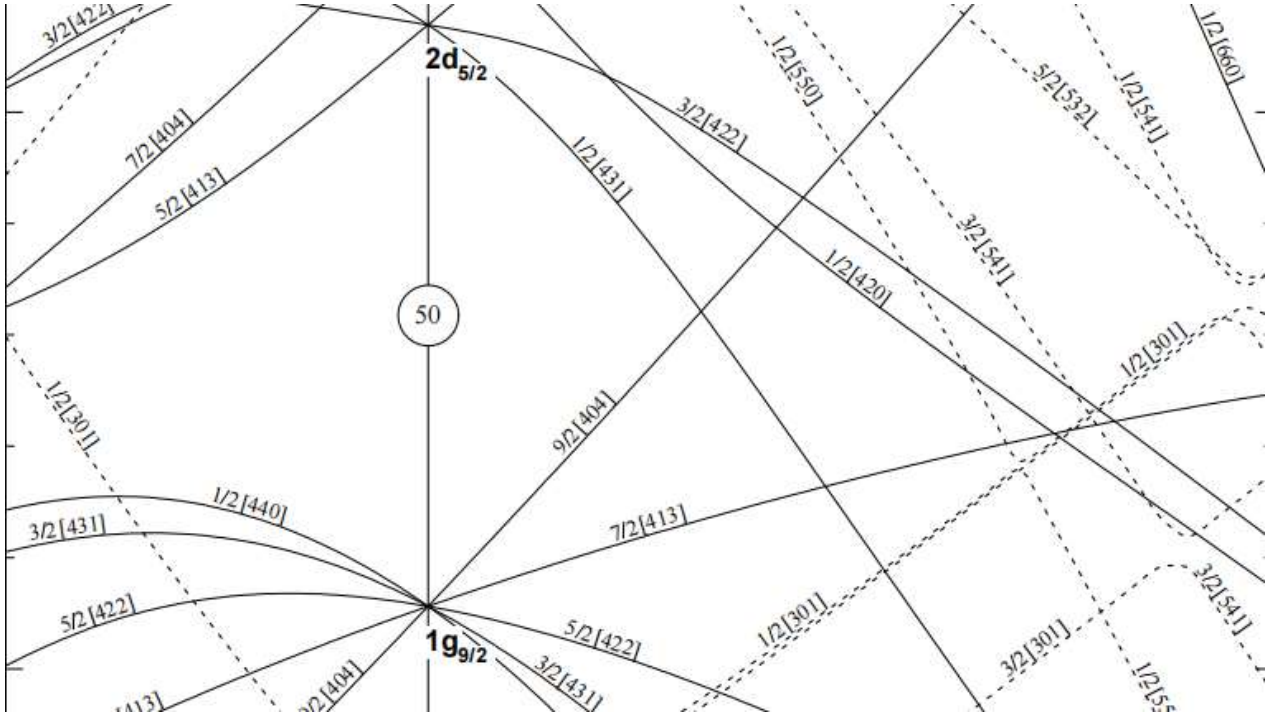
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Introduction

What is an Island of Inversion?

- Strong quadrupole correlations overcome the spherical shell-gap cost, allowing intruder configurations to dominate the ground state.
- N=20:$1d_{3/2}$, $1f_{7/2}$, $2p_{3/2}$
- N=40:$1f_{5/2}$, $1g_{9/2}$, $2d_{5/2}$





Experimental motivation around N=40

$^{68}Ni \rightarrow ^{66}Fe \rightarrow ^{64}Cr$

- The high-lying 2^+ state observed in ^{68}Ni and its low B(E2) value are the result of the relatively large energy gap separating the pf and $1g_{9/2}$ orbitals.
- ^{66}Fe, with only two protons less, shows a sudden change in nuclear structure with an increased collectivity manifested via its very low-lying 2^+ state.
- ^{64}Cr is the nucleus where the theoretical calculations predicted the lowest-lying 2^+ level in the region.

Theoretical Framework

Model space and LNPS interaction

- Model space(^{48}Ca as core)

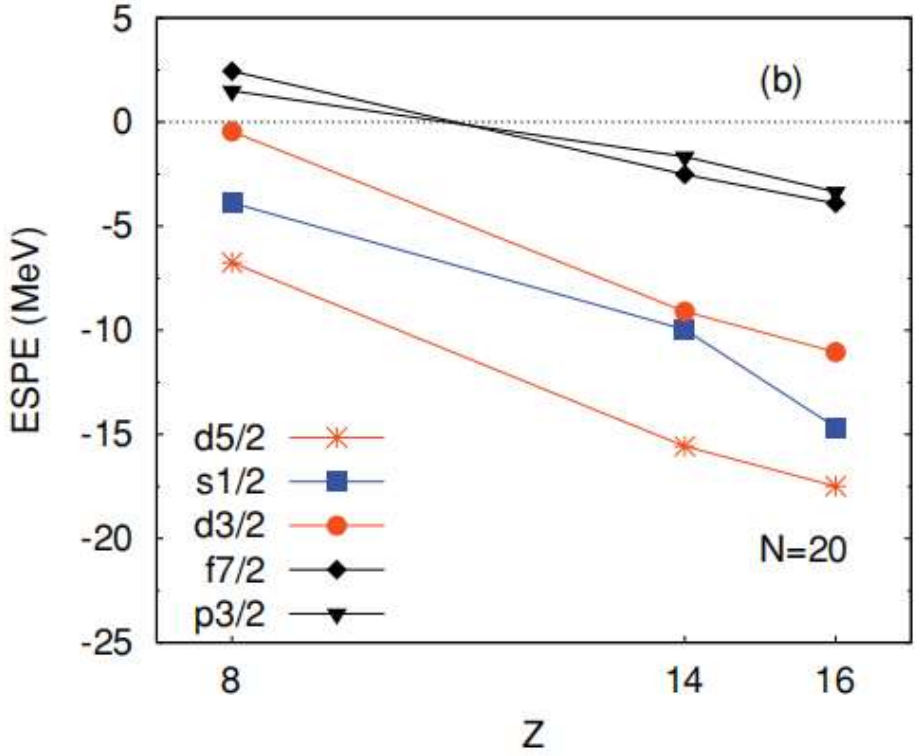
Proton: $1f_{7/2}$,$2p_{3/2}$,$1f_{5/2}$,$2p_{1/2}$

Neutron: $2p_{3/2}$,$2p_{1/2}$,$1f_{5/2}$,$1g_{9/2}$,$2d_{5/2}$

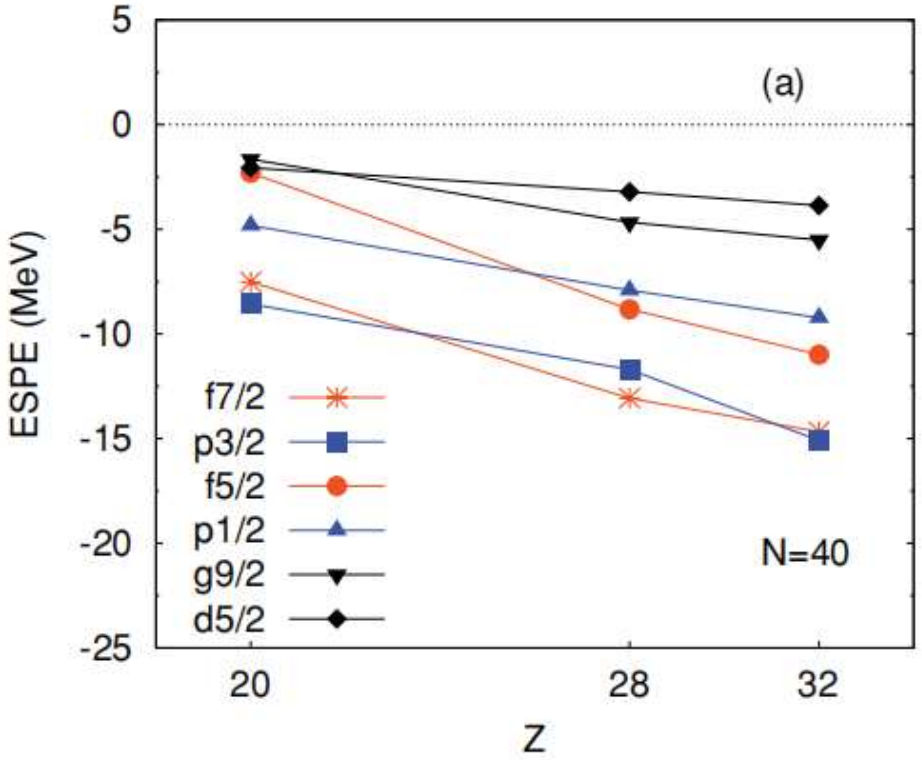
- Interaction

The LNPS interaction is a hybrid shell-model interaction designed for the N≈40 region, combining realistic two-body matrix elements with monopole corrections and explicitly including the neutron $1g_{9/2}$ and $2d_{5/2}$ intruder orbitals.

Shell evolution and the analogy between N=20 and N=40



- $1d_{3/2}$ - $1f_{7/2}$ gap ↓
- $1f_{7/2}$ - $2p_{3/2}$ quadrupole partners

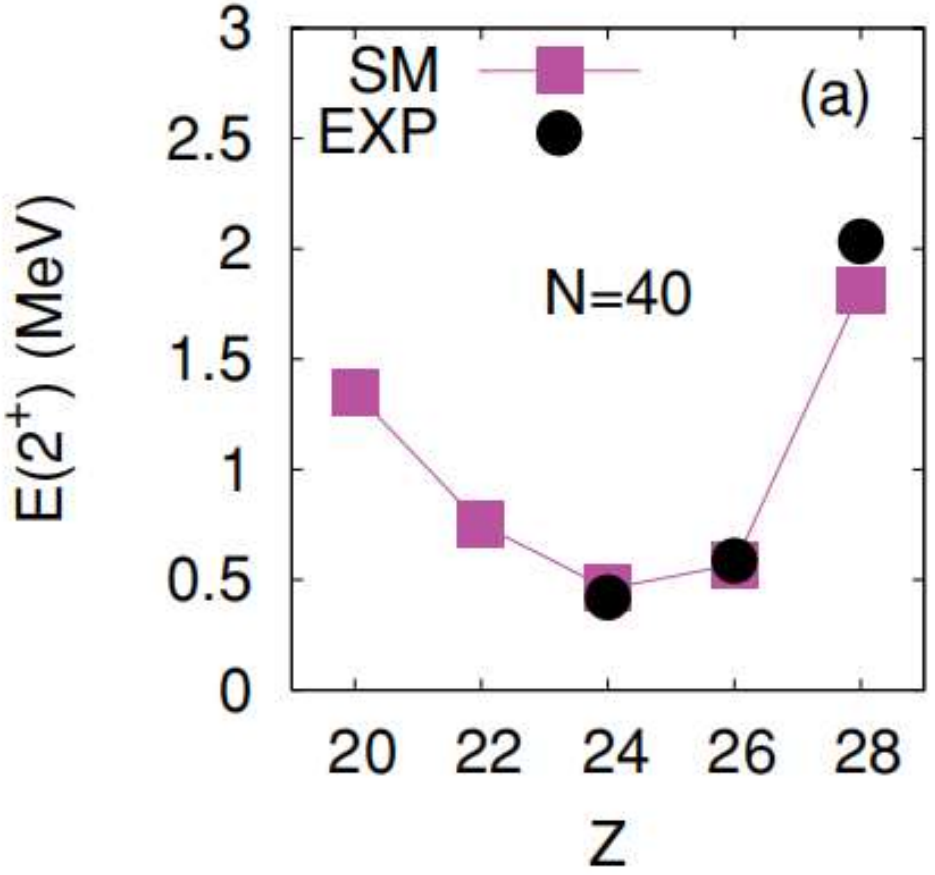


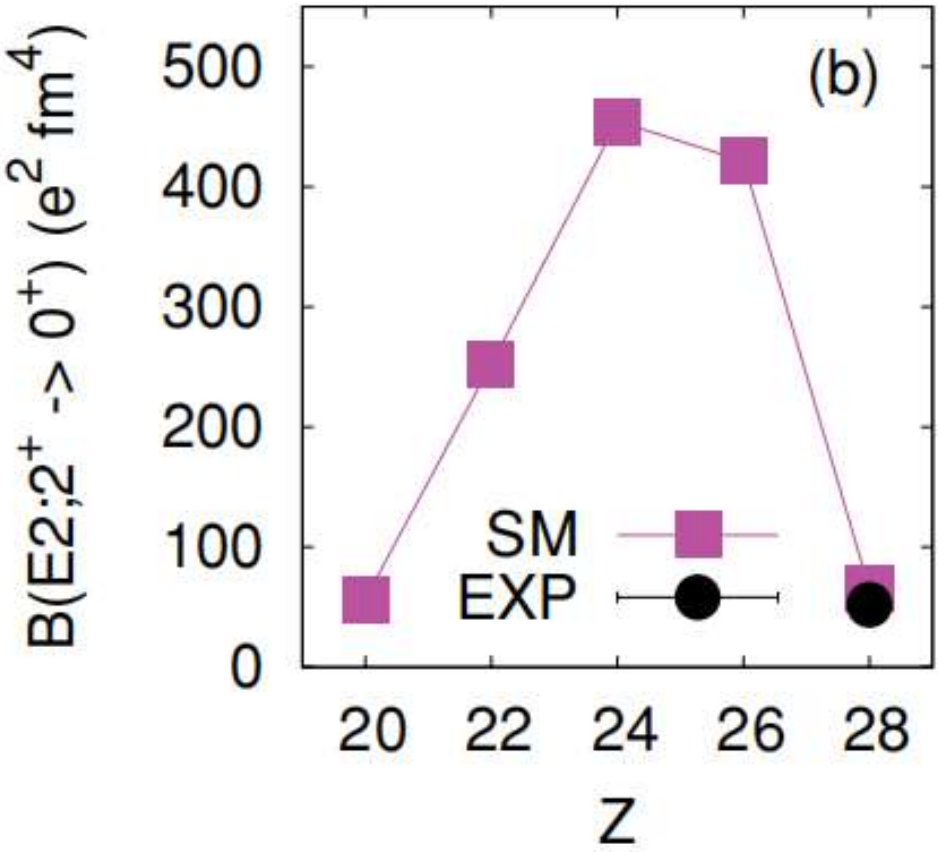
- $1f5_{/2}$ - $1g9_{/2}$ gap ↓
- $1g9_{/2}$ - $2d_{35/2}$ quadrupole partners

Lower excitation cost + strong quadrupole correlations

Results

Where is the maximum collectivity along N=40 ?

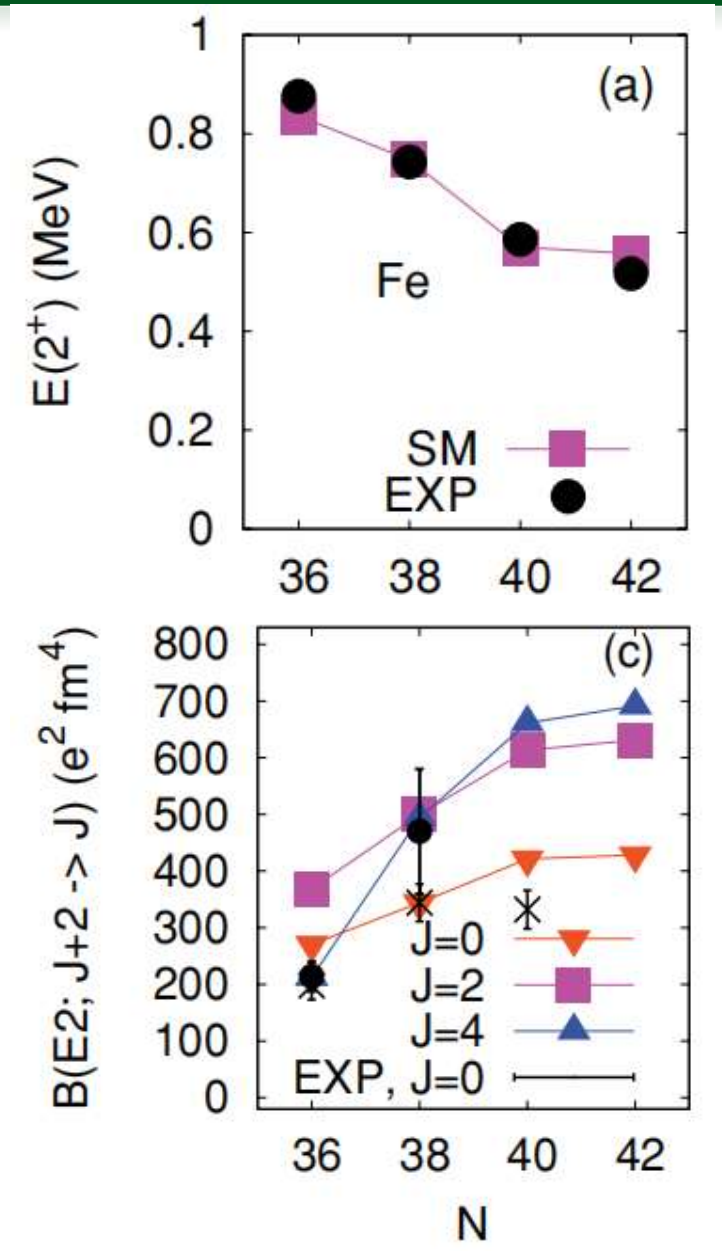


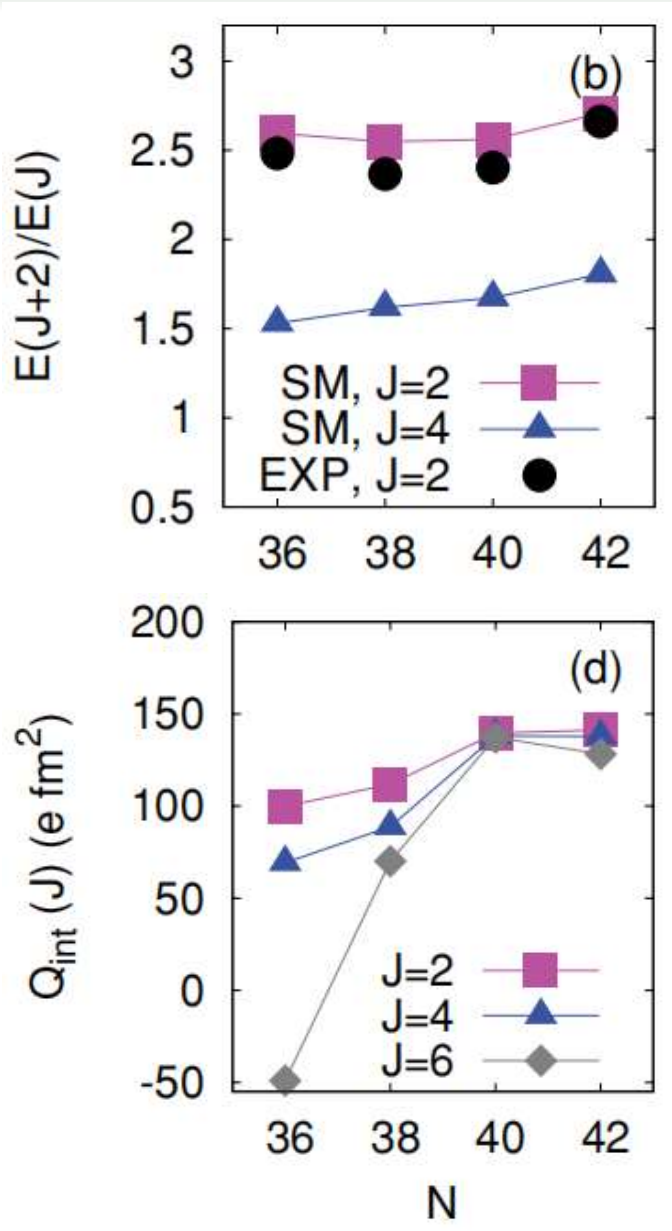


Microscopic structure

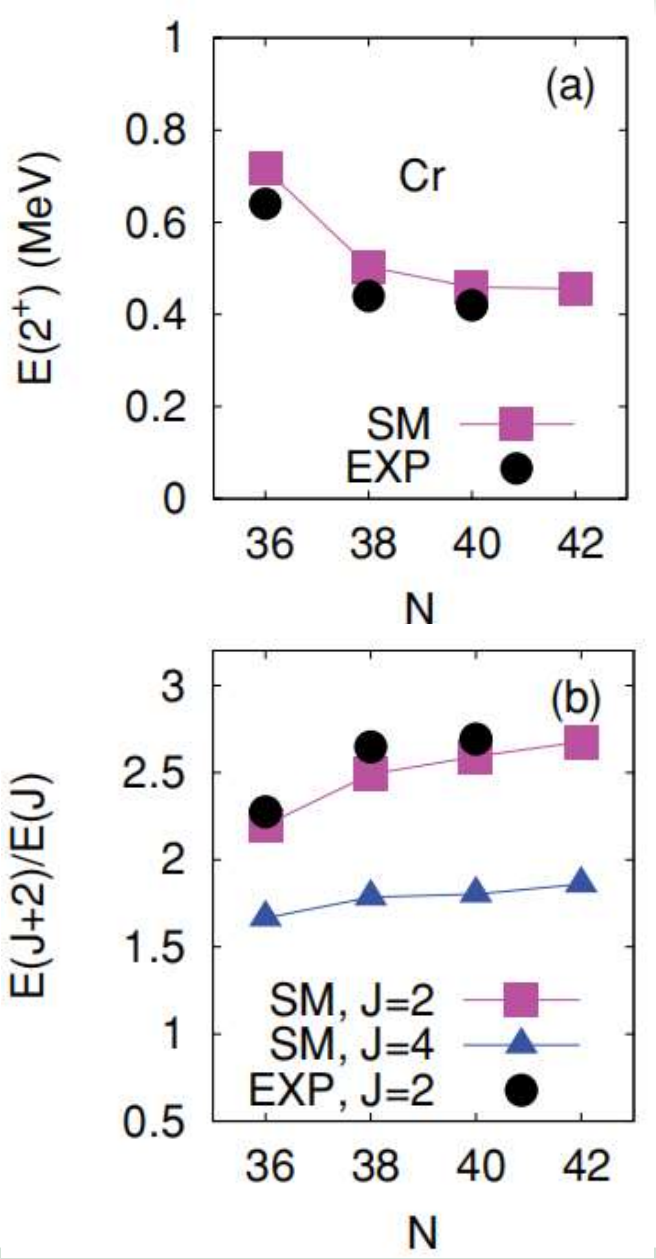
Nucleus	$\nu g_{9/2}$	$\nu d_{5/2}$	0p0h	2p2h	4p4h	6p6h	E_{corr}
^{68}Ni	0.98	0.10	55.5	35.5	8.5	0.5	−9.03
^{66}Fe	3.17	0.46	1	19	72	8	−23.96
^{64}Cr	3.41	0.76	0	9	73	18	−24.83
^{62}Ti	3.17	1.09	1	14	63	22	−19.62
^{60}Ca	2.55	1.52	1	18	59	22	−12.09

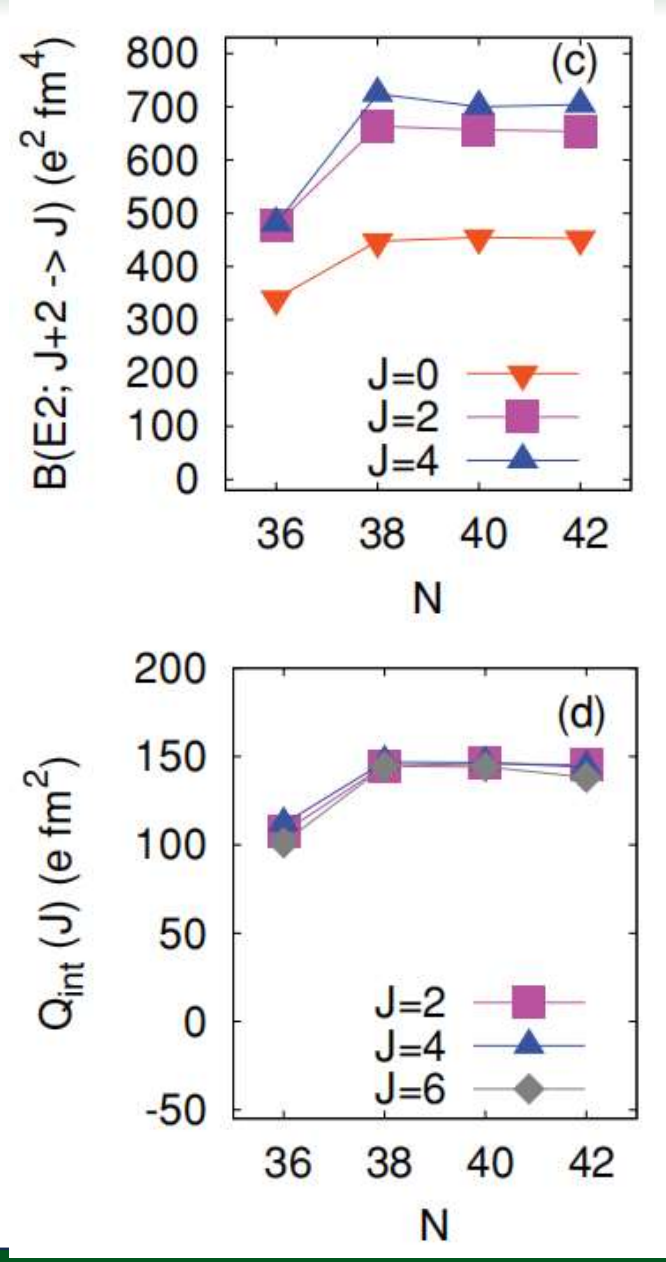
Results of Fe isotopes





Results of Cr isotopes

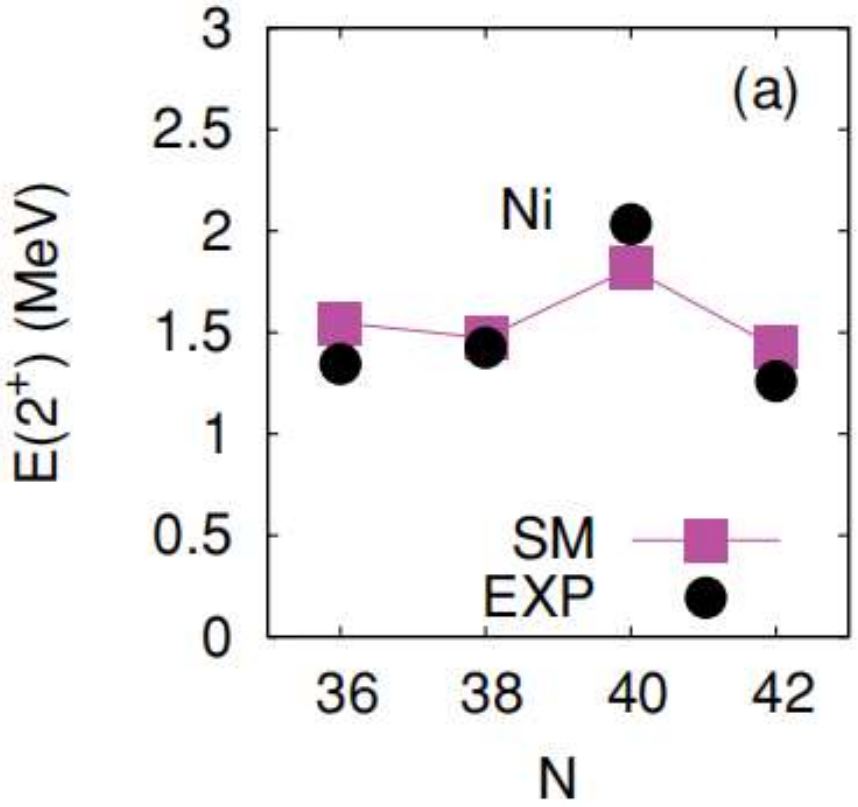


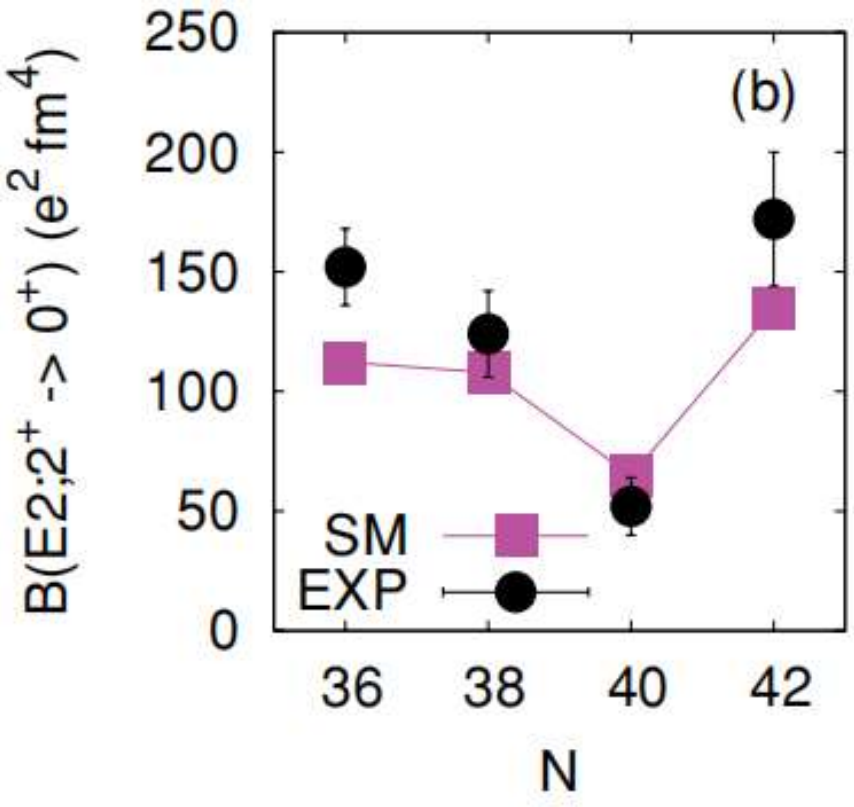


Intruder occupations along Fe and Cr

Nucleus	N	$\nu 0g_{9/2}$	$\nu 1d_{5/2}$
^{62}Fe	36	0.95	0.12
^{64}Fe	38	2.0	0.27
^{66}Fe	40	3.22	0.51
^{68}Fe	42	2.30	0.62
^{60}Cr	36	1.55	0.31
^{62}Cr	38	2.77	0.66
^{64}Cr	40	3.41	0.76
^{66}Cr	42	2.28	0.90

Results of Ni isotopes





Conclusions

Conclusions

- Shell evolution

$1f_{5/2}$–$1g_{9/2}$ gap decreases below ^{68}Ni.

- Microscopic mechanism

$1g_{9/2}$–$2d_{5/2}$ intruder configurations + strong p-n quadrupole correlations

- Different deformation onset

Cr:N=38 Fe:N=40

- Center of the island

^{64}Cr exhibits the maximum collectivity and deformation, showing features characteristic of a good rotor

Thank you for attention !